PROJECT TITLE

STOCK MARKET PRICE PREDICTION & ANALYSIS USING PYTHON

ABSTRACT

The stock market is one of the most complex and volatile financial systems in the world, driven by macroeconomic indicators, investor sentiment, corporate earnings, and geopolitical events. Accurate prediction of stock price movements provides a significant competitive advantage to investors, portfolio managers, and financial institutions. This project presents a comprehensive data analytics and machine learning pipeline in Python to analyze and predict stock prices using historical OHLCV (Open, High, Low, Close, Volume) data sourced from Yahoo Finance.

The project covers the complete data science workflow: data collection via the yfinance API rigorous preprocessing, feature engineering of technical indicators (Moving Averages, RSI, MACD, Bollinger Bands), Exploratory Data Analysis (EDA), and multi-chart visualization using Matplotlib and Seaborn. A Long Short-Term Memory (LSTM) neural network — a deep learning architecture specifically suited for time-series forecasting — was designed and trained to predict next-day closing prices. The model achieved a Root Mean Squared Error (RMSE) of 393.86 INR on a 60-day test set, with a Mean Absolute Percentage Error (MAPE) of 1.39%, indicating high predictive accuracy.

This project demonstrates proficiency in Python, Pandas, NumPy, Keras/TensorFlow, Scikit-learn, Matplotlib, and Seaborn — a skill set directly aligned with roles in quantitative analysis, data science, and financial technology (FinTech). The insights derived are relevant to real-world algorithmic trading and portfolio optimization applications.

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